

Elektor Zener Diode Tester Kit

Construction Manual



SKU 21289

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This manual is available as a downloadable PDF.
You can find this essential document at the following address:
www.elektor.com/21289

Notice

This document is complementary to the information contained in the engineering background and discussions posted on the Elektor Labs website.

Web links to these publications may be found in Section 10.

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1. Kit Contents

The kit includes the PCB and all parts listed in the Bill of Materials in section 8.

The next photo shows what components to expect in the kit. The parts shown were collected for the prototype. Some parts may look slightly different, depending on the manufacturer of the parts used for the kit.

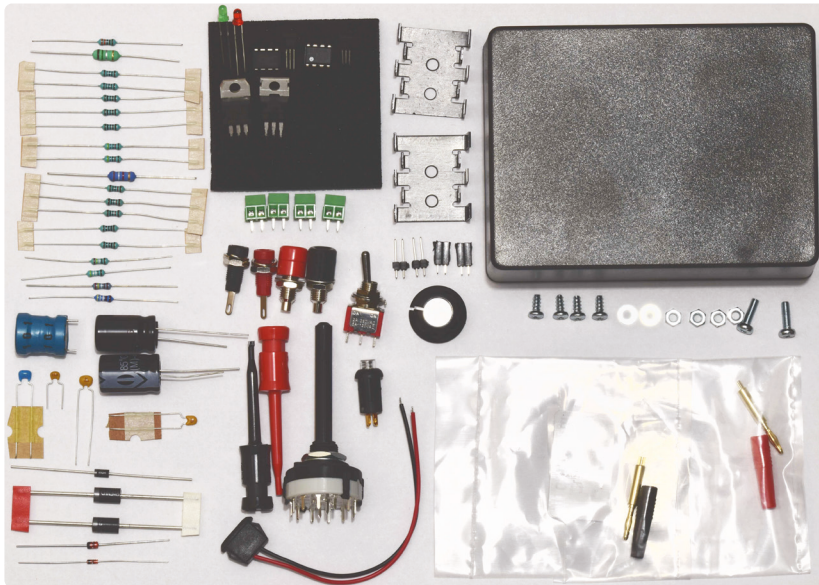


Fig. 1. Components in the Elektor Zener Diode Tester Kit (except the wires).

The following figure shows the top overlay of the PCB in the kit.

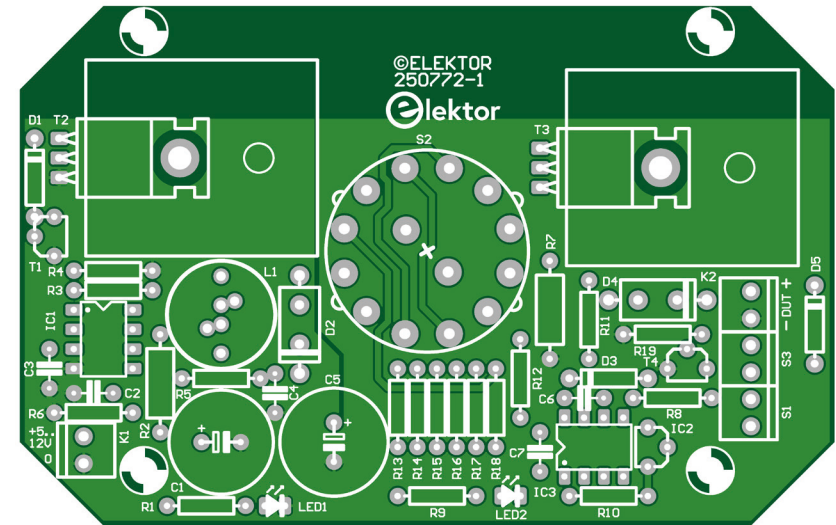


Fig. 2. PCB design of the Zener Diode Tester 250772 v1.1.

2. Tools Needed

- › Soldering equipment for through-hole components
- › Cutting pliers
- › Lead bending tool for through-hole resistors
- › Flat-jaw pliers (to bend the leads of TO-220 transistors)
- › Pozidriv and Phillips screwdriver
- › Tool for hex screws (Allen wrench), HEX = 1.5 mm (depending on the screw in the knob)
- › PC or laptop and color printer (to print the front sticker)
- › Scissors and a sharp hobby knife to cut the front sticker(s) to size

3. Assembling the PCB

Summary: Start by soldering the lowest parts first and the highest ones last. Then fasten the two power transistors to their heat sinks and mount the assemblies onto the PCB. Next the rotary switch S2 and place IC1 and IC2 into the sockets (observe the correct orientation).

Recommended order:

D5
R1, R3-R6, R8-R19
D3
D1
R2, R7
IC sockets
D2, D4
C2, C3
C4, C6, C7
Screw terminal blocks K1, K2, S1, S3

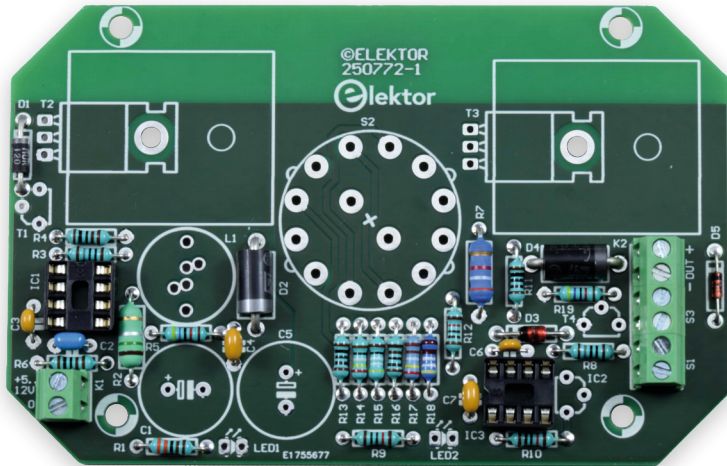


Fig. 3. Resistors, diodes, and screw terminal blocks soldered (PCB 250772-1, v1.0).

Next

T1, T4, and IC2 (all have TO-92 packages)
2-way pin headers for LED1 and LED2
L1
C5
C1

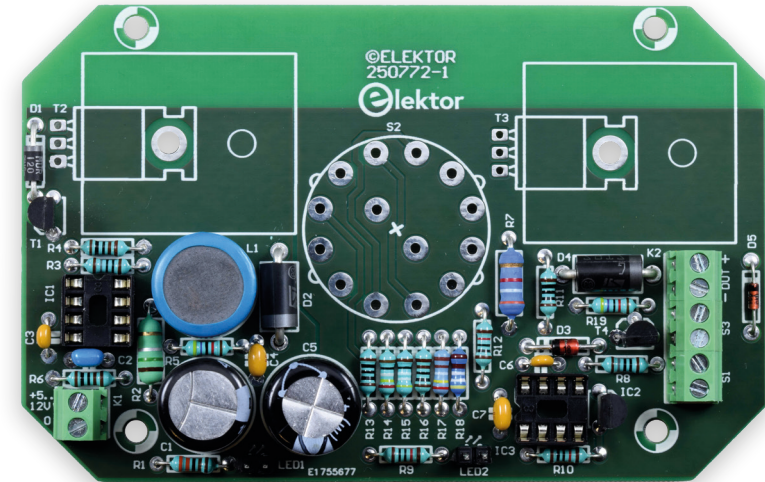


Fig. 4. T1, T4, IC2, pin headers, L1, C5, and C1 soldered (PCB 250772-1, v1.0).

Next

Apply a thin layer of thermal grease on the back of T2 and T3. No insulation is needed; the transistors can be fitted directly onto the heat sinks. First, a screw and a nut are used to fasten each transistor to the heat sink. Temporarily place the screw of the heat sinks in the mounting hole on the PCB to see where the leads of the transistors must be bent perpendicularly. There is some room for error here. Bend the leads. Place the heat sinks on the PCB; the leads of the transistors should fit smoothly through the designated pads, and fasten each with a second nut on the bottom side. **Do not place the heat sinks against the PCB, this can cause short circuits!** The copper plane on top is connected to ground but has large clearances where the two screws from the heat sinks are, so the nut on the top side cannot make contact. The screws are connected to the drain of the MOSFETs. Only then, solder the leads of T2 and T3.

Place rotary switch S2 and solder all its pins. Be sure it is placed fully flat against the PCB!
Put IC1 and IC3 in the IC sockets.



Fig. 5. Heat sinks fixed to the PCB, T2, T3, and S2 soldered, IC1 and IC3 placed in their sockets.

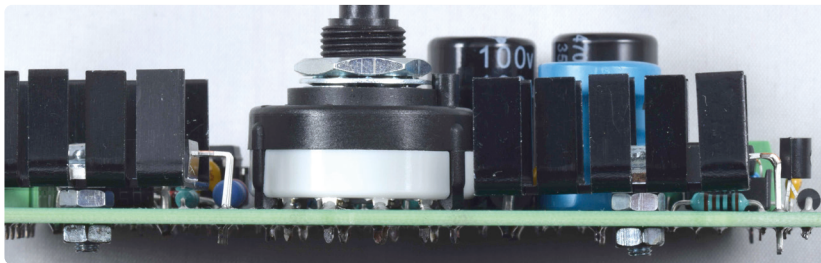


Fig. 6. Mounting of the heat sinks, side view.

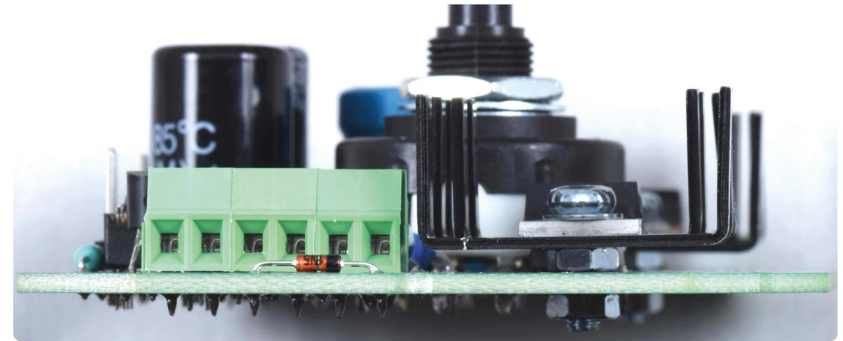


Fig. 7. Side view of the mounting of the heat sink of T3. The heat sink of T2 is visible in the back.

The PCB is now ready for testing. Connect the mA inputs of a multimeter directly to the output (K2) and measure the six test currents. Also measure the +60 VDC supply voltage at a 50 mA output current, on the cathode of D2 with respect to ground. Be careful not to touch anything while testing!

4. Construction

A PDF of the front design with dimensions is available on our Labs website (see Chapter 10 for the web link, look for Project Elements – Other).

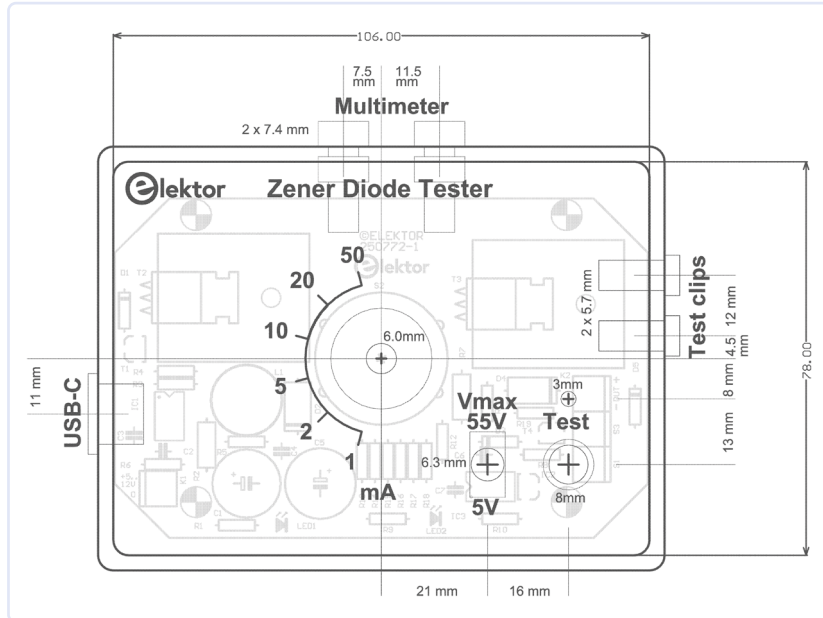


Fig. 8. Dimensions of the front design with respect to the PCB and enclosure. All are referenced to the spindle of the rotary switch in the center of the front.

The size of the design is 106 × 78 mm, and a rounded rectangle marks the borders (center of the lines). In **Figure 8**, the sides of the enclosure are also shown as the outer rounded rectangle. Print it on normal paper and make sure the borders have the correct dimensions. Cut the inner rounded rectangle with scissors from the paper, and cut the round 6-mm hole for the spindle of the rotary switch with a sharp hobby knife. The location of the spindle of the rotary switch is the only critical location; it is in the exact center of the enclosure. The PCB is designed this way.

Inside the enclosure, there is a small circle, also in the center, which can be helpful. This can also be used to drill a hole in the correct position for the 6-mm spindle of the rotary switch first. Use adhesive tape to temporarily fasten it to the top side of the enclosure. Place it in the exact center of the front; the holes for the spindle should now align. Create the other three holes in the top of the enclosure:

6.3 mm for the toggle switch S3

8 mm for pushbutton S1

3 mm for LED2: use a 2.5 mm drill first, and use a small round file to make just big enough for the LED to stick. Using a 3 mm drill could result in a hole that is a little too big for it to stay in place. Use a small round file to make it bigger and test several times if the LED will fit through and be fixed properly.

For the other holes, start by making smaller holes with, for instance, a 2 mm drill, possibly by hand. The plastic enclosure is very easy to work on, or maybe too easy to work on. Be patient when making the holes; the ABS plastic is soft, and a hole can be made too large without applying much force.

When the holes in the top of the enclosure are ready, drill the holes for the 2-mm banana sockets with a 4.5 mm drill. The 6 mm thread (approx. 5.7 mm wide) has two flat sides that are less than 5 mm wide. Use a small half-round file to slowly make the connectors fit and align properly.

The two holes for the two 4-mm banana connectors are round and should be 7.4 mm in diameter.

Also make a slotted hole for the USB-C socket to fit through. Do not make it too wide, or it will be pulled out when a USB connector is plugged in.

The center of the holes in the sides is 18 mm from the top of the enclosure.

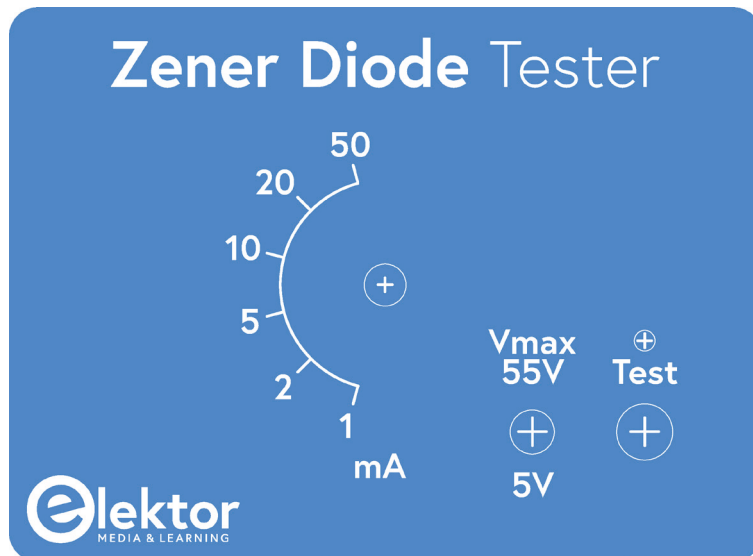


Fig. 9. Front design with logo and corporate identity, color, and typography.

A PDF of this front design is available on our Labs website (see Chapter 10 for the web link, look for Project Elements – Other). Print it on a self-adhesive (glossy) photo paper and make sure the size is correct, exactly the same as printed before on normal paper, 106 × 78 mm. A cheaper paper can act a little like blotting paper when printing in high quality. Printing in standard quality was fine. The matte version was worse than the glossy paper. Test different printer settings.

Stick the front design in the center of the enclosure, exactly like the normal paper before, and make sure it is perfectly aligned with all the holes, so that the scale aligns with the rotary switch and the text with the two switches. Cut out the paper that covers the four holes carefully with a sharp hobby knife. Fit the two switches and the 3-mm LED. Tighten the nut of the toggle switch very carefully, making sure not to damage the front. Shorten the leads of the LED, but leave the short lead slightly shorter to still indicate the cathode. Use a drop of superglue to make sure the LED will never fall back inside the enclosure. You may also secure the USB-C connector with a drop of superglue.

5. Wiring



Fig. 10. Front sticker (earlier design), connectors and switches fitted, and PCB fixed to the bottom of the enclosure.

In **Figure 10**, a sticker with an earlier design is fixed to the front. All the connectors are mounted, and the PCB is fixed to the bottom of the enclosure by four #4-1/4" self-tapping screws.

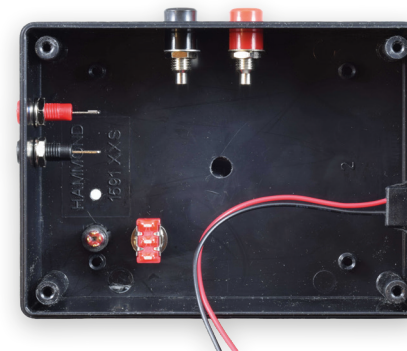


Fig. 11. Inside view of the enclosure with connectors and switches fitted.

Next, place the top of the enclosure on its side. Place the bottom right next to it. Now everything can be wired with 0.25 mm² (24 AWG) wire. The following is a list of a logical order and lengths of the pieces of wire needed. On pushbutton S1, the solder lug of the cathode of the LED is marked with a grey spot. Place it toward the side of the enclosure. The solder lug on the opposite side is the anode. The two other solder lugs, to the left and right, are for the switch. Place the 2-way pin sockets on

the 2-way pin headers of the LEDs. Twist the wires of each pair before soldering.

Connect (from PCB)	Color	Length [cm]	Comment
S3 to toggle switch	red	10	
	black	10	
LED1 to pushbutton	black	12	cathode only
S1 to pushbutton	red	10	
	black	10	
LED1 to pushbutton	red	12	anode
LED2 to green LED	red	10	anode
	black	10	cathode
4 mm banana to	red	9	red to red connector
2 mm banana	black	6	black to black connector
+ of DUT (K2)	red	12	to red 2 mm banana connector
- of DUT (K2)	black	12	to black 2 mm banana connector

In total, 63 cm red wire is needed and 60 cm black wire.



Fig. 12. Everything is connected.

Check all wiring. When placing the top of the enclosure on the bottom, make sure the wires are not touching the heatsinks, just to be cautious. In normal use, they don't get hot. Next, shorten the spindle of the rotary switch S2. Remove about 11 mm, maybe a little more. The plastic is surprisingly tough. When placing the knob over the spindle, it shouldn't be able to touch the front, as the printed front layout is vulnerable. It is best to do this before starting the wiring. Clamp the top of the spindle in a bench vise and use a saw to remove 11 mm of the spindle. Place the top of the enclosure on the bottom and check if enough has been removed. The knob of the prototype is secured with a headless Allen screw. A 1.5-mm Allen key is needed to fix this knob. When this is done, the Zener Diode Tester is ready for use.



Fig. 13. Top view of the finished Zener Diode Tester.



Fig. 14. Side views of the finished Zener Diode Tester.

6. In Practice

The test clips can each be connected with 30 cm 0.25 mm² (24 AWG) wire to the 2-mm banana plugs. This length should be long enough. To measure SMD components and similar devices, where the test clips cannot be attached, standard measuring cables with test tips for 4-mm banana plugs can be connected to the 4-mm banana sockets on the enclosure. Most measuring cables can be connected in parallel, as their plugs also have a 4-mm socket. A multimeter can then be connected in parallel to the measuring cables and to the DUT.

7. Replacing the Sticker

Since the sticker is printed with an inkjet printer, the front can get damaged easily and become covered with scratches over time. Remove the knob and open the enclosure by removing the four screws on the bottom. Remove the sockets of the LEDs from the 2-way pin headers. Remove the wires from the pushbutton S1 from its screw terminal. Remove the nut from the toggle switch and, from the inside, pull it out of the front. The pushbutton can be pushed out of the front from the inside. The wires will fit through the hole.

Remove the old sticker by pulling it very slowly off the front. Remove residual glue and paper with sticker remover and make sure the front is dry and clean. Place the top back on the bottom, the spindle of the rotary switch is now protruding through the front. After printing a new sticker, cut it from the sheet of self-adhesive photopaper. Cut the round 6-mm hole for the rotary switch and the hole for the LED, the latter hole slightly smaller than 3 mm. Peel off and cut off a part of the protective paper from the sticker with a pair of scissors at the side where the LED is, about 2.5 cm. Place the sticker over the spindle and make sure the hole for the LED is exactly placed over the LED, then push the sticker onto the front.

Take the top of the enclosure off the bottom and remove the rest of the protective paper from the back of the sticker, then push it carefully and evenly onto the front. Cut the holes for the two switches with a sharp hobby knife and place the switches back. Connect all the wires and close the enclosure. Place the knob on the spindle, and the tester is as good as new.

8. Bill of Materials

Resistors

R1 = 3.3 k Ω , 600 mW, 1 %

R2 = 0.15 Ω , 1 W, 5 %, lead spacing 12.7 mm, diam. 3.5 mm

R3, R8 = 180 Ω , 600 mW, 1 %

R4, R6, R16 = 1 k Ω , 600 mW, 1 %

R5, R19 = 47 k Ω , 600 mW, 1 %

R7 = 3.9 k Ω , 1 W, 5 %, lead spacing 12.7 mm, diam. 3.5 mm

R9, R10, R12 = 10 k Ω , 600 mW, 1 %

R11, R13 = 100 Ω , 600 mW, 1 %

R14 = 249 Ω , 600 mW, 1 %

R15 = 499 Ω , 600 mW, 1 %

R17 = 2.49 k Ω , 250 mW, 1 %

R18 = 4.99 k Ω , 250 mW, 1 %

Capacitors

C1 = 470 μ F, 35 V, 20 %, I_r = 1.86 A, D = 12.5 mm, LS = 5 mm,
ESR = 0.038 Ω (Panasonic EEUTP1V471)

C2 = 1 μ F, 100 V, 10 %, cer. X7R, LS 5 mm

C3 = 2.2 nF, 50 V, 10 %, cer. X7R, LS 5 mm

C4, C7 = 100 nF, 100 V, 10 %, cer. X7R, LS 5 mm

C5 = 100 μ F, 100 V, 20%, D 14 mm max., LS 5 mm

C6 = 1 nF, 100 V, 10 %, cer. X7R, LS 5 mm

Inductor

L1 = 100 μ H, Irms 2.4 A, Isat 3.5 A. 0.09 Ω , radial, pitch 2.5/5/10 mm,
Diam. 14 mm max. (Kemet SBC8-101-242)

Semiconductors

D1 = MUR120G, DO-41

D2, D4 = STPS3150RL, DO-201AD

D3 = Zener diode, 18 V, 1.3 W (BZX85C18)

D5 = Zener diode, 5.1 V, 0.5 W (1N5231B)

LED1 = LED in pushbutton S1

LED2 = LED, green, T-1 (3 mm)

IC1 = MC34063, DIP-8

IC2 = LM4040BIZ-5.0/NOPB, TO-92 (TO-226AA-3)

IC3 = TL051CP, DIP-8

T1 = BC327.25, TO-92

T2 = STP40NF10L, TO-220

T3 = IRF9510PBF, TO-220

T4 = 2N7000, TO-92

Miscellaneous

K1, K2, S1, S3 = 2-way screw terminal block, LS 3.5 mm, max. 1.5 mm²
connected to K1 = USB-C socket, 30 V / 3 A, chassis-mounted, wired,
9 x 16 mm

connected to K2 = Banana Connector, 2 mm, Socket, 60 VDC,
Panel Mount, Red (24.102.1 Multicomp)

connected to K2 = Banana Connector, 2 mm, Socket, 60 VDC,
Panel Mount, Black (24.102.2 Multicomp)

Test clip, red, 30 VAC/60 VDC (931467101 Hirschmann)

Test clip, black, 30 VAC/60 VDC (931467100 Hirschmann)

S1 (= LED1) = Pushbutton, panel mount, SPST-NO, 30 VDC, 100 mA,
red illuminated

S2 = Rotary Switch, 6 Position, 2 Pole, 30 °, 150 mA, 250 V (CK1050 Lorlin)

S3 = Toggle switch, panel mount, SPDT, solder lugs, 28 VDC

T2, T3 = Heat sink 15°C/W (FK 214 SA-CB Fischer Elektronik)

Enclosure Hammond 1591XXSSBK, ABS, 110x82x44 mm

S2 = Knob, Round Shaft 6 mm, Diam. 21 mm, with indicator line
(CP-LB21-6-6D Multicomp)

4 x self-tapping screw #4-1/4", carbon steel, pan head

Banana plug, red, 2 mm, 60 VDC (25.205.1 Multicomp)

Banana plug, black, 2 mm, 60 VDC (25.205.2 Multicomp)

4 mm banana socket, panel mount, black (24.247.2 Multicomp)

4 mm banana socket, panel mount, red (24.247.1 Multicomp)

LED1, LED2 = Pin header, 1x2, vertical, pitch 2.54 mm

LED1, LED2 = Pin socket, 1x2, vertical, pitch 2.54 mm

T2, T3 = M3 screw, 10 mm, steel, pan head

2 x M3 washer, plain, steel

4 x M3 nut

IC1, IC3 = DIP-8 socket

Black wire, 0.25 mm² (24 AWG), stranded 14 x 0.15 mm, 1 m

Red wire, 0.25 mm² (24 AWG), stranded 14 x 0.15 mm, 1 m

9. Specifications and Properties

Supply voltage	5–12 V/2 A (USB-C, see text)
Supply current ($I_{test} = 50 \text{ mA}$)	1.1 A (5 V) 412 mA (12 V)
Selectable output currents (fixed)	1, 2, 5, 10, 20, 50 mA
Maximum test voltage	> 54 V (when I_{test} is correct)
Maximum no load output voltage	60 V ($I_{test} \ll$ value selected)
5 V limit voltage	4.5 V @ $I_{test} = 1 \text{ mA}$ 5.18 V @ $I_{test} = 50 \text{ mA}$
Dimensions enclosure	110 x 82 x 44 mm
Size PCB	105.4 x 67.3 mm
4 mm banana sockets to connect a multimeter	
2 mm banana sockets to connect test clips	
USB-C socket to connect a power supply	
PCB designed for use with Hammond 1591XXSSBK ABS enclosure (specifically position of mounting holes)	

10. Web Links

Zener Diode Tester kit:

www.elektor.com/21289



Zener Diode Tester project on the Elektor Labs website:

www.elektormagazine.com/labs/zenner-diode-tester-250772



11. Schematic and PCB Layout

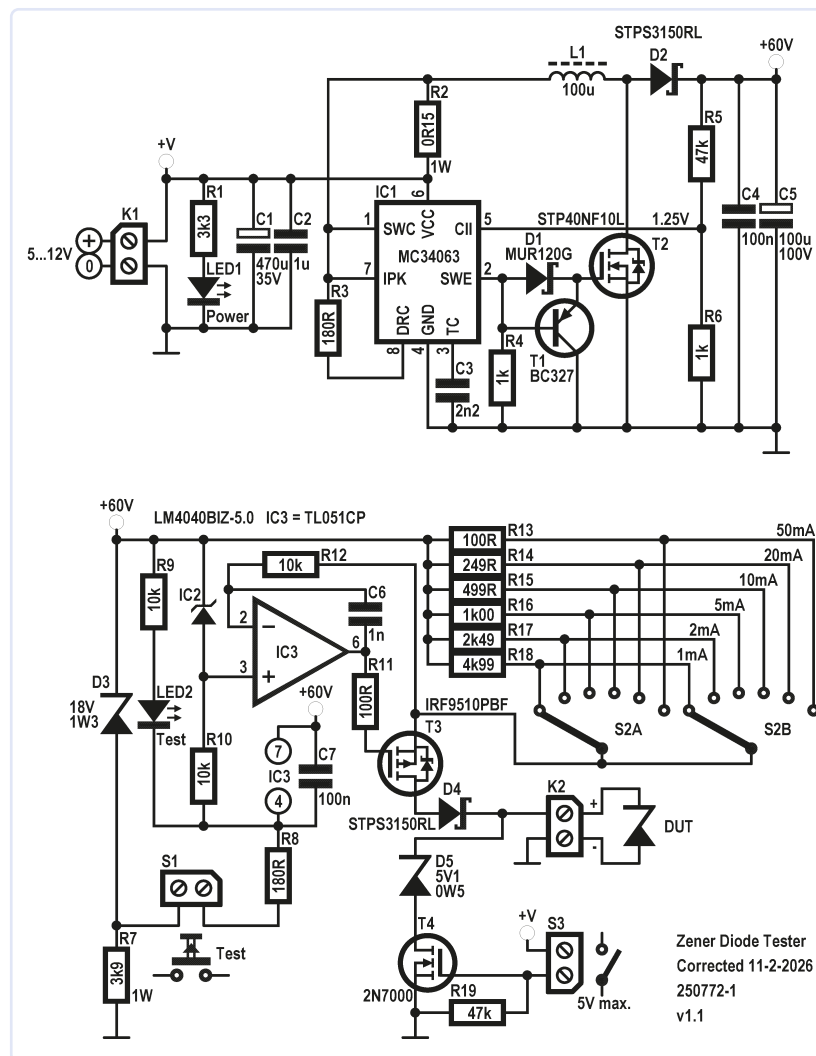


Fig. 15. Schematic of the Elektor Zener Diode Tester

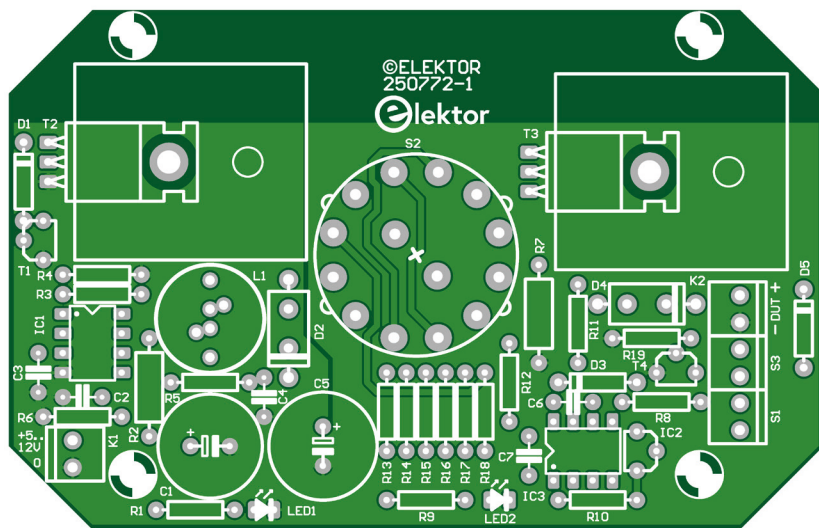


Fig. 16. Top overlay of the PCB for the Zener Diode Tester (250772-1, v1.1).

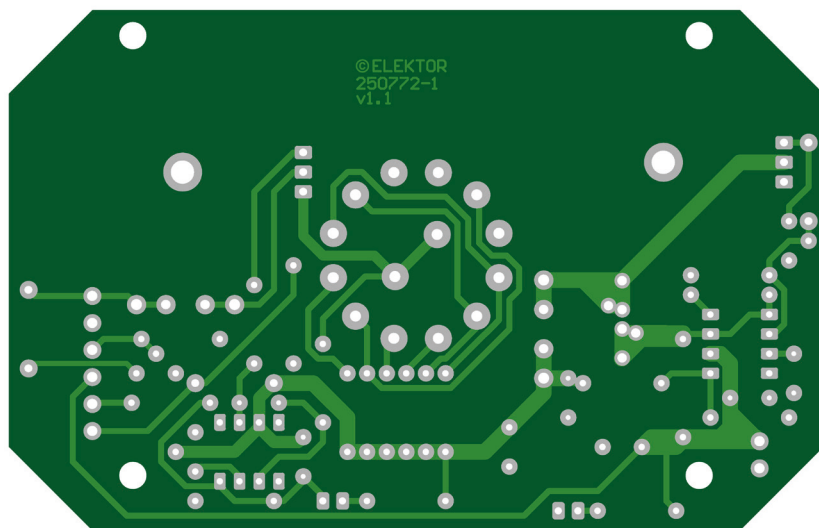


Fig. 17. Bottom overlay of the PCB of the Zener Diode Tester (250772-1, v1.1).

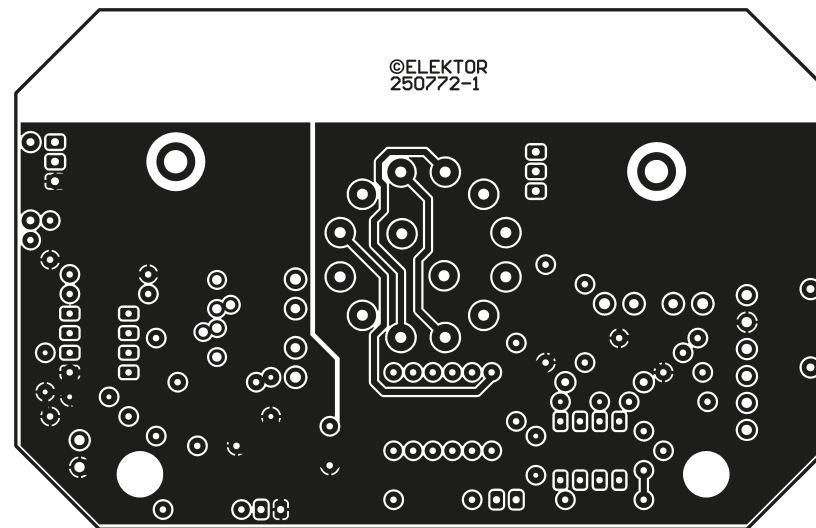


Fig. 18. Copper on top of the PCB for the Zener Diode Tester (250772-1, v1.1).

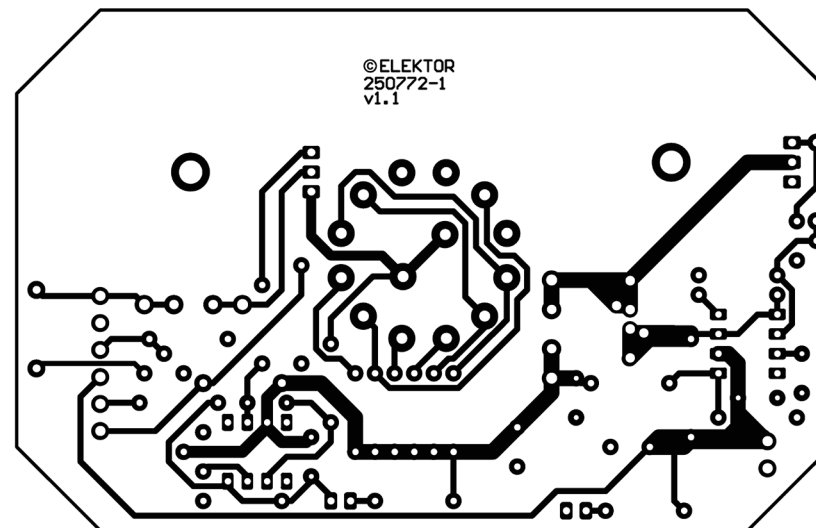


Fig. 19. Copper on the bottom of the PCB of the Zener Diode Tester (250772-1, v1.1).